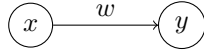


1 Twice example

1.1 Model



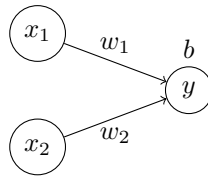
$$y = x \cdot w$$

1.2 Cost

$$\begin{aligned} C &= (xw - y)^2 \\ \frac{dC}{dw} &= \frac{d}{dw}(xw - y)^2 \\ &= 2(xw - y)x \end{aligned}$$

2 Gates example

2.1 Model



$$\begin{aligned} y &= \sigma(x_1w_1 + x_2w_2 + b) \\ \sigma(x) &= \frac{1}{1 + e^{-x}} \\ \frac{d}{dx}\sigma(x) &= \sigma(x)(1 - \sigma(x)) \end{aligned}$$

2.2 Cost

$$a = \sigma(x_1w_1 + x_2w_2 + b)$$

$$\begin{aligned} \frac{\partial a}{\partial w_1} &= \frac{\partial}{\partial w_1}\sigma(x_1w_1 + x_2w_2 + b) \\ &= \sigma(x_1w_1 + x_2w_2 + b)(1 - \sigma(x_1w_1 + x_2w_2 + b)) \frac{\partial}{\partial w_1}(x_1w_1 + x_2w_2 + b) \\ &= a(1 - a)x_1 \end{aligned}$$

$$\begin{aligned}
\frac{\partial a}{\partial w_2} &= \frac{\partial}{\partial w_2} \sigma(x_1 w_1 + x_2 w_2 + b) \\
&= \sigma(x_1 w_1 + x_2 w_2 + b)(1 - \sigma(x_1 w_1 + x_2 w_2 + b)) \frac{\partial}{\partial w_2} (x_1 w_1 + x_2 w_2 + b) \\
&= a(1 - a)x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a}{\partial b} &= \frac{\partial}{\partial b} \sigma(x_1 w_1 + x_2 w_2 + b) \\
&= \sigma(x_1 w_1 + x_2 w_2 + b)(1 - \sigma(x_1 w_1 + x_2 w_2 + b)) \frac{\partial}{\partial b} (x_1 w_1 + x_2 w_2 + b) \\
&= a(1 - a)
\end{aligned}$$

$$C = (a_i - y_i)^2$$

$$\begin{aligned}
\frac{\partial C}{\partial w_1} &= \frac{\partial}{\partial w_1} (a - y)^2 \\
&= 2(a - y) \frac{\partial}{\partial w_1} (a - y) \\
&= 2(a - y)a(1 - a)x_1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_2} &= \frac{\partial}{\partial w_2} (a - y)^2 \\
&= 2(a - y) \frac{\partial}{\partial w_2} (a - y) \\
&= 2(a - y)a(1 - a)x_2
\end{aligned}$$

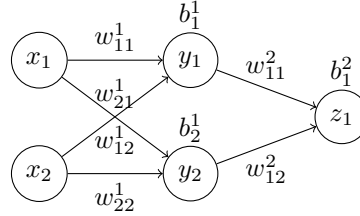
$$\begin{aligned}
\frac{\partial C}{\partial b} &= \frac{\partial}{\partial b} (a - y)^2 \\
&= 2(a - y) \frac{\partial}{\partial b} (a - y) \\
&= 2(a - y)a(1 - a)
\end{aligned}$$

2.3 Cost - Simplified

$$\begin{aligned}
 a &= \sigma(x_1 w_1 + x_2 w_2 + b) \\
 \frac{\partial a}{\partial w_i} &= a(1 - a)x_i \\
 \frac{\partial a}{\partial b} &= a(1 - a) \\
 C &= (a - y)^2 \\
 \frac{\partial C}{\partial w_i} &= 2(a - y)a(1 - a)x_i \\
 \frac{\partial C}{\partial b} &= 2(a - y)a(1 - a)
 \end{aligned}$$

3 Xor example

3.1 Model



$$\begin{aligned}
 y_1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
 y_2 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
 z_1 &= \sigma(y_1 w_{11}^2 + y_2 w_{12}^2 + b_1^2) \\
 \sigma(x) &= \frac{1}{1 + e^{-x}} \\
 \frac{d}{dx} \sigma(x) &= \sigma(x)(1 - \sigma(x))
 \end{aligned}$$

3.2 Cost

$$\begin{aligned}
 a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
 a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
 a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial a_1^1}{\partial w_{11}^1} &= \frac{\partial}{\partial w_{11}^1} \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1 - \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)) \frac{\partial}{\partial w_{11}^1} (x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
 &= a_1^1(1 - a_1^1)x_1
 \end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^1}{\partial w_{12}^1} &= \frac{\partial}{\partial w_{12}^1} \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1 - \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)) \frac{\partial}{\partial w_{12}^1} (x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= a_1^1(1 - a_1^1)x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_2^1}{\partial w_{21}^1} &= \frac{\partial}{\partial w_{21}^1} \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)) \frac{\partial}{\partial w_{21}^1} (x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)x_1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_2^1}{\partial w_{22}^1} &= \frac{\partial}{\partial w_{22}^1} \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)) \frac{\partial a_2^1}{\partial w_{22}^1} \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^1}{\partial b_1^1} &= \frac{\partial}{\partial b_1^1} \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)(1 - \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1)) \frac{\partial}{\partial b_1^1} (x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
&= a_1^1(1 - a_1^1)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_2^1}{\partial b_2^1} &= \frac{\partial}{\partial b_2^1} \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)(1 - \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1)) \frac{\partial}{\partial b_2^1} (x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
&= a_2^1(1 - a_2^1)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{11}^2} &= \frac{\partial}{\partial w_{11}^2} \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)(1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{11}^2} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2(1 - a_1^2)a_1^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{12}^2} &= \frac{\partial}{\partial w_{12}^2} \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{12}^2} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_2^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial b_1^2} &= \frac{\partial}{\partial b_1^2} \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial b_1^2} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{11}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{11}^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_1 w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{12}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{12}^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_2 w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{21}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{21}^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_1 w_{12}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial w_{22}^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial w_{22}^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_2 w_{12}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial b_1^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial b_1^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial b_2^1} &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial b_2^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) w_{12}^2
\end{aligned}$$

$$C = (a_1^2 - z_1)^2$$

$$\begin{aligned}\frac{\partial C}{\partial w_{11}^2} &= \frac{\partial}{\partial w_{11}^2} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{11}^2} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1\end{aligned}$$

$$\begin{aligned}\frac{\partial C}{\partial w_{12}^2} &= \frac{\partial}{\partial w_{12}^2} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{12}^2} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1\end{aligned}$$

$$\begin{aligned}\frac{\partial C}{\partial b_1^2} &= \frac{\partial}{\partial w_{12}^2} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{12}^2} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2)\end{aligned}$$

$$\begin{aligned}\frac{\partial C}{\partial w_{11}^1} &= \frac{\partial}{\partial w_{11}^1} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{11}^1} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_1 w_{11}^2\end{aligned}$$

$$\begin{aligned}\frac{\partial C}{\partial w_{12}^1} &= \frac{\partial}{\partial w_{12}^1} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{12}^1} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_2 w_{11}^2\end{aligned}$$

$$\begin{aligned}\frac{\partial C}{\partial w_{21}^1} &= \frac{\partial}{\partial w_{21}^1} (a_1^2 - z_1)^2 \\ &= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{21}^1} (a_1^2 - z_1) \\ &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_1 w_{12}^2\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{22}^1} &= \frac{\partial}{\partial w_{22}^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial w_{22}^1} (a_1^2 - z_1) \\
&= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_2 w_{12}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_1^1} &= \frac{\partial}{\partial b_1^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial b_1^1} (a_1^2 - z_1) \\
&= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_2^1} &= \frac{\partial}{\partial b_2^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial b_2^1} (a_1^2 - z_1) \\
&= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) w_{12}^2
\end{aligned}$$

3.3 Cost - Simplified

$$\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
C &= (a_1^2 - z_1)^2 \\
\frac{\partial C}{\partial w_{11}^2} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 \\
\frac{\partial C}{\partial w_{12}^2} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 \\
\frac{\partial C}{\partial b_1^2} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) \\
\frac{\partial C}{\partial w_{11}^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_1 w_{11}^2 \\
\frac{\partial C}{\partial w_{12}^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) x_2 w_{11}^2 \\
\frac{\partial C}{\partial w_{21}^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_1 w_{12}^2 \\
\frac{\partial C}{\partial w_{22}^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) x_2 w_{12}^2 \\
\frac{\partial C}{\partial b_1^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 (1 - a_1^1) w_{11}^2 \\
\frac{\partial C}{\partial b_2^1} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 (1 - a_2^1) w_{12}^2
\end{aligned}$$

3.4 Backpropagation

$$\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial a_1^1} &= \frac{\partial}{\partial a_1^1} \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial a_1^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial a_1^2}{\partial a_2^1} &= \frac{\partial}{\partial a_2^1} \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) (1 - \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2)) \frac{\partial}{\partial a_2^1} (a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
&= a_1^2 (1 - a_1^2) w_{12}^2
\end{aligned}$$

$$C = (a_1^2 - z_1)^2$$

$$\begin{aligned}
\frac{\partial C}{\partial a_1^2} &= \frac{\partial}{\partial a_1^2} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial a_1^2} (a_1^2 - z_1) \\
&= 2(a_1^2 - z_1)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial a_1^1} &= \frac{\partial}{\partial a_1^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial a_1^1} (a_1^2 - z_1) \\
&= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) w_{11}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial a_2^1} &= \frac{\partial}{\partial a_2^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) \frac{\partial}{\partial a_2^1} (a_1^2 - z_1)^2 \\
&= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) w_{12}^2 \\
&= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) w_{12}^2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{11}^2} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_1^1 \\
&= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) a_1^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{12}^2} &= 2(a_1^2 - z_1) a_1^2 (1 - a_1^2) a_2^1 \\
&= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) a_2^1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_1^2} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2) \\
&= \frac{\partial C}{\partial a_1^2}a_1^2(1 - a_1^2)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{11}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_1w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1}a_1^1(1 - a_1^1)x_1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{12}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)x_2w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1}a_1^1(1 - a_1^1)x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{21}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_1w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1}a_2^1(1 - a_2^1)x_1
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial w_{22}^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)x_2w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1}a_2^1(1 - a_2^1)x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_1^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_1^1(1 - a_1^1)w_{11}^2 \\
&= \frac{\partial C}{\partial a_1^1}a_1^1(1 - a_1^1)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial C}{\partial b_2^1} &= 2(a_1^2 - z_1)a_1^2(1 - a_1^2)a_2^1(1 - a_2^1)w_{12}^2 \\
&= \frac{\partial C}{\partial a_2^1}a_2^1(1 - a_2^1)
\end{aligned}$$

3.5 Backpropagation - Simplified

$$\begin{aligned}
a_1^1 &= \sigma(x_1 w_{11}^1 + x_2 w_{12}^1 + b_1^1) \\
a_2^1 &= \sigma(x_1 w_{21}^1 + x_2 w_{22}^1 + b_2^1) \\
a_1^2 &= \sigma(a_1^1 w_{11}^2 + a_2^1 w_{12}^2 + b_1^2) \\
C &= (a_1^2 - z_1)^2 \\
\frac{\partial C}{\partial a_1^2} &= 2(a_1^2 - z_1) \\
\frac{\partial C}{\partial a_1^1} &= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) w_{11}^2 \\
\frac{\partial C}{\partial a_2^1} &= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) w_{12}^2 \\
\frac{\partial C}{\partial w_{11}^2} &= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) a_1^1 \\
\frac{\partial C}{\partial w_{12}^2} &= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) a_2^1 \\
\frac{\partial C}{\partial b_1^2} &= \frac{\partial C}{\partial a_1^2} a_1^2 (1 - a_1^2) \\
\frac{\partial C}{\partial w_{11}^1} &= \frac{\partial C}{\partial a_1^1} a_1^1 (1 - a_1^1) x_1 \\
\frac{\partial C}{\partial w_{12}^1} &= \frac{\partial C}{\partial a_1^1} a_1^1 (1 - a_1^1) x_2 \\
\frac{\partial C}{\partial w_{21}^1} &= \frac{\partial C}{\partial a_2^1} a_2^1 (1 - a_2^1) x_1 \\
\frac{\partial C}{\partial w_{22}^1} &= \frac{\partial C}{\partial a_2^1} a_2^1 (1 - a_2^1) x_2 \\
\frac{\partial C}{\partial b_1^1} &= \frac{\partial C}{\partial a_1^1} a_1^1 (1 - a_1^1) \\
\frac{\partial C}{\partial b_2^1} &= \frac{\partial C}{\partial a_2^1} a_2^1 (1 - a_2^1)
\end{aligned}$$

3.6 Backpropagation - Generalised

$$\begin{aligned}
\frac{\partial C}{\partial a_k^{(l-1)}} &= \sum_j \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)} (1 - a_j^{(l)}) w_{jk}^{(l)} \\
\frac{\partial C}{\partial w_{jk}^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)} (1 - a_j^{(l)}) a_k^{(l-1)} \\
\frac{\partial C}{\partial b_j^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} a_j^{(l)} (1 - a_j^{(l)})
\end{aligned}$$

4 Backpropagation

4.1 Column

$$a^{(l)} = \begin{bmatrix} a_0^{(l)} \\ a_1^{(l)} \\ \vdots \\ a_n^{(l)} \end{bmatrix} \quad b^{(l)} = \begin{bmatrix} b_0^{(l)} \\ b_1^{(l)} \\ \vdots \\ b_n^{(l)} \end{bmatrix} \quad w^{(l)} = \begin{bmatrix} w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1n}^{(l)} \\ w_{21}^{(l)} & w_{22}^{(l)} & \cdots & w_{2n}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1}^{(l)} & w_{m2}^{(l)} & \cdots & w_{mn}^{(l)} \end{bmatrix}$$

$$a^{(0)} = x$$

$$z^{(l+1)} = a^{(l)} \cdot w^{(l)} + b^{(l)}$$

$$a^{(l)} = \sigma(z^{(l)})$$

$$C = (a^{(l)} - y)^2$$

$$\frac{\partial C}{\partial a_j^{(L)}} = 2(a_j^{(L)} - y_j)$$

$$\frac{\partial C}{\partial a_k^{(l-1)}} = \sum_j \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)}) w_{jk}^{(l)}$$

$$\frac{\partial C}{\partial w_{jk}^{(l)}} = \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)}) a_k^{(l-1)}$$

$$\frac{\partial C}{\partial b_j^{(l)}} = \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)})$$

4.2 Row

$$a^{(l)} = \begin{bmatrix} a_0^{(l)} & a_1^{(l)} & \cdots & a_m^{(l)} \end{bmatrix} \quad b^{(l)} = \begin{bmatrix} b_0^{(l)} & b_1^{(l)} & \cdots & b_m^{(l)} \end{bmatrix}$$

$$w^{(l)} = \begin{bmatrix} w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1n}^{(l)} \\ w_{21}^{(l)} & w_{22}^{(l)} & \cdots & w_{2n}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m1}^{(l)} & w_{m2}^{(l)} & \cdots & w_{mn}^{(l)} \end{bmatrix}$$

$$a^{(0)} = x$$

$$z^{(l+1)} = a^{(l)} \cdot w^{(l)} + b^{(l)}$$

$$a^{(l)} = \sigma(z^{(l)})$$

$$\begin{aligned}
C &= (a^{(l)} - y)^2 \\
\frac{\partial C}{\partial a_j^{(L)}} &= 2(a_j^{(L)} - y_j) \\
\frac{\partial C}{\partial a_k^{(l-1)}} &= \sum_j \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)}) w_{kj}^{(l)} \\
\frac{\partial C}{\partial w_{kj}^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)}) a_k^{(l-1)} \\
\frac{\partial C}{\partial b_j^{(l)}} &= \frac{\partial C}{\partial a_j^{(l)}} \sigma'(z_j^{(l)})
\end{aligned}$$